Supplementary Material S2: Search Protocol and Record Flow

Recursive Training on Model-Generated Data: A Critical Integrative Review

of Failure Regimes, Adversarial Interactions and Governance

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A1. Sources, strings and record flow

All searches were run on 29 October 2026 for records published from January 2022. The date filter was applied in the query for OpenAlex (`from_publication_date`) and Scopus (`PUBYEAR > 2021`), and after retrieval for arXiv, OpenReview and IEEE Xplore. Duplicates were identified first by arXiv identifier or DOI and then by normalised title (Unicode NFKD, ASCII folding, case and punctuation removed). Table 1 gives the record flow; the search scripts are available at the project repository; the exact query strings as executed are listed below.

Table 1: Record flow of the structured search.

Source	Retrieved	Outside window	In window
OpenAlex	797	0	797
Scopus	594	0	594
arXiv	630	54	576
OpenReview	543	78	465
IEEE Xplore	95	29	66
Total	2,659	161	2,498
Duplicates removed			865
Unique records screened (title/abstract)			1,633
Full texts assessed			664
Evidence studies included via the structured search			10

Recursive-training terms (TERMS). "model collapse" | "knowledge collapse" | "self-consuming" | "model autophagy" | "curse of recursion" | autophagous | "recursive training" | "iterative retraining" | "self-trained" | "data feedback loop(s)" | "closed-loop learning" | "self-correcting loop" | "iterated learning" | "self-rewarding" | "iterative DPO" | "iterative preference".

Generative-model terms (GEN). "generative model" | "language model" | LLM | "diffusion model" | "foundation model" | "synthetic data" | "synthetic text" | "generated data".

Per-source syntax. *OpenAlex*: TERMS AND GEN in title and abstract; `from_publication_date:2` cursor-paginated. *Scopus*: `TITLE-ABS-KEY(TERMS) AND TITLE-ABS-KEY(GEN)`; wildcard `self-train*`; `PUBYEAR > 2021`; institutional access CAPES/UFPA. *arXiv*: each TERMS phrase in `ti:` or `abs:`; GEN required in `abs:`; categories cs.LG, cs.CL, cs.CR, cs.AI; 54 pre-2022 records excluded after retrieval. *OpenReview*: 13 phrase queries; the query list omits "model autophagy", "curse of recursion", "autophagous" and "self-trained" because the OpenReview interface does not support them reliably; year derived from a truncated Unix timestamp, decoded post-retrieval. *IEEE Xplore*: 11 paired queries (a TERMS phrase paired with a GEN phrase where the interface required it); date filter post-retrieval.

Semantic Scholar: not searched owing to API rate limits at the cutoff date.

A2. List of Included Sources

The manuscript cites 101 sources in total: 61 evidence studies, 8 position paper(s), 23 context sources, and 9 legal and normative instruments (Supplementary Table S1 (S1_cited_sources.csv)). Their category (E evidence, P position, C context, L legal/normative), entry route (S structured search, A anchor, CH citation chasing, R added during revision to address a reviewer comment or to verify a claim), anchor status, and preprint/workshop flag (†) are provided in Supplementary Table S1 (S1_cited_sources.csv). Evidence studies (N=61) are those whose formal results, experiments or empirical observations are directly synthesised in the body of the review; position, context and legal/normative sources are cited for arguments, background or regulatory reference. Category reflects how each source is used in this review.